

Forecasting how much water a reservoir will hold, and what state it will be in, months ahead

An integrated hydrological-limnological framework producing seasonal (1–7 month) predictions of reservoir **quantity** and **quality**, built to inform adaptive water management under a drying Mediterranean climate.

Daniel Mercado-Bettín¹ · Rafael Marcé¹

¹ Centre d'Estudis Avançats de Blanes (CEAB-CSIC), Blanes, Catalonia, Spain
daniel.mercado@csic.es

See the forecasts running on the pilot reservoirs →



LIVE FORECAST

• The management problem

Reservoir operators are being asked to make decisions on a timescale that no existing tool covers well.

Prolonged droughts, intense rainfall and shifting seasonality alter hydrological regimes and biogeochemical processes **at the same time**. Storage falls while thermal stratification strengthens, oxygen is stripped from the hypolimnion and algal biomass responds, each change compounding the others.

Weather forecasts reach days. Climate projections describe decades. The **seasonal gap in between** is exactly where supply planning, drought declarations and abstraction decisions are actually made.

THE FORECASTING GAP

Weather forecasts
hours – 10 days

This framework
1 – 7 months

Climate projections
decades

• What the framework predicts

A single chain generates both families of variables from the same climate forcing, so quantity and quality forecasts stay physically consistent with one another.

WATER QUANTITY

- **System storage** Sau + Susqueda combined volume
- **Inflows** six gauges across the Ter and Tordera

WATER QUALITY

- **Outlet gate temperature** seven gates across both reservoirs
- **River temperature** six river reaches
- **Water residence time** AED2 age tracer, per gate
- **Saline intrusion risk** 52 coastal wells

In development: nutrients, dissolved oxygen and chlorophyll *a*.

• Pilot: the Ter system

The framework is being applied to **Sau** and **Susqueda**, the tandem reservoirs on the River Ter that supply drinking water to the Barcelona metropolitan area, a system pushed to record lows during the 2021–2024 Catalan drought.

Both are managed by the **Catalan Water Agency (ACA)**, which sets the operational questions the forecasts are designed to answer: how much water will arrive, how low will storage go, and what condition will that water be in when it gets there.

READING EVERY FORECAST BELOW

- **Above normal**
upper tercile (> P66)
- **Normal**
middle tercile (P33–P66)
- **Below normal**
lower tercile: the scarcity signal (< P33)

Terciles come from the observed record, so “normal” means normal *for that calendar month*.

• From a climate ensemble to an operational decision

The chain runs unattended on the 15th of every month. Every stage carries all 51 members forward, so each output is a **probability distribution**, never a single number.

FORCING

SEAS5.1 seasonal ensemble

51 members from the Copernicus CDS, re-downloaded every month.

CORRECTION

Bias adjustment & downscaling

Every member corrected against the ERA5 reanalysis record.

MODEL 1

Catchment hydrology

Rainfall-runoff over the Ter sub-basins → inflows and system storage.

MODEL 2

GLM 3.1.1 + AED2

1-D thermal structure and mixing at Sau and Susqueda, forced by those inflows.

PRODUCT

Published every month

Tercile probabilities and verified skill, pushed to the live viewers.

QUANTITY OUT system storage · inflows

QUALITY OUT gate temperature · residence time · saline intrusion

a The Ter system



b Modelled outlet gates

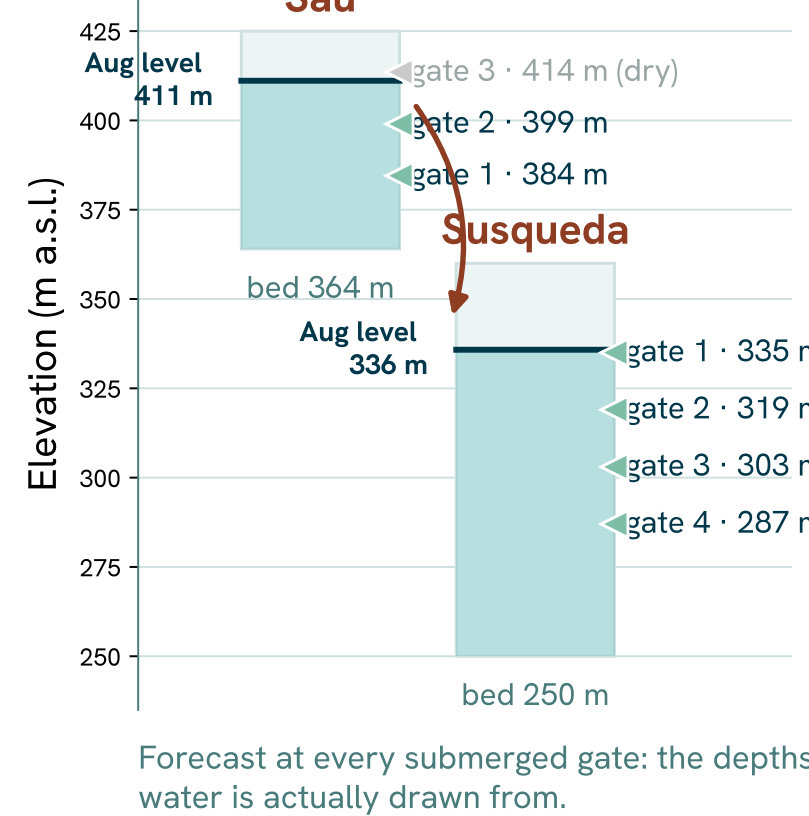


Fig. 1 The modelled Ter catchment, with Sau and Susqueda in series on the river. The gates are the depths each reservoir actually releases and abstracts water from.

Water quality: forecast temperature at the four Susqueda outlet gates

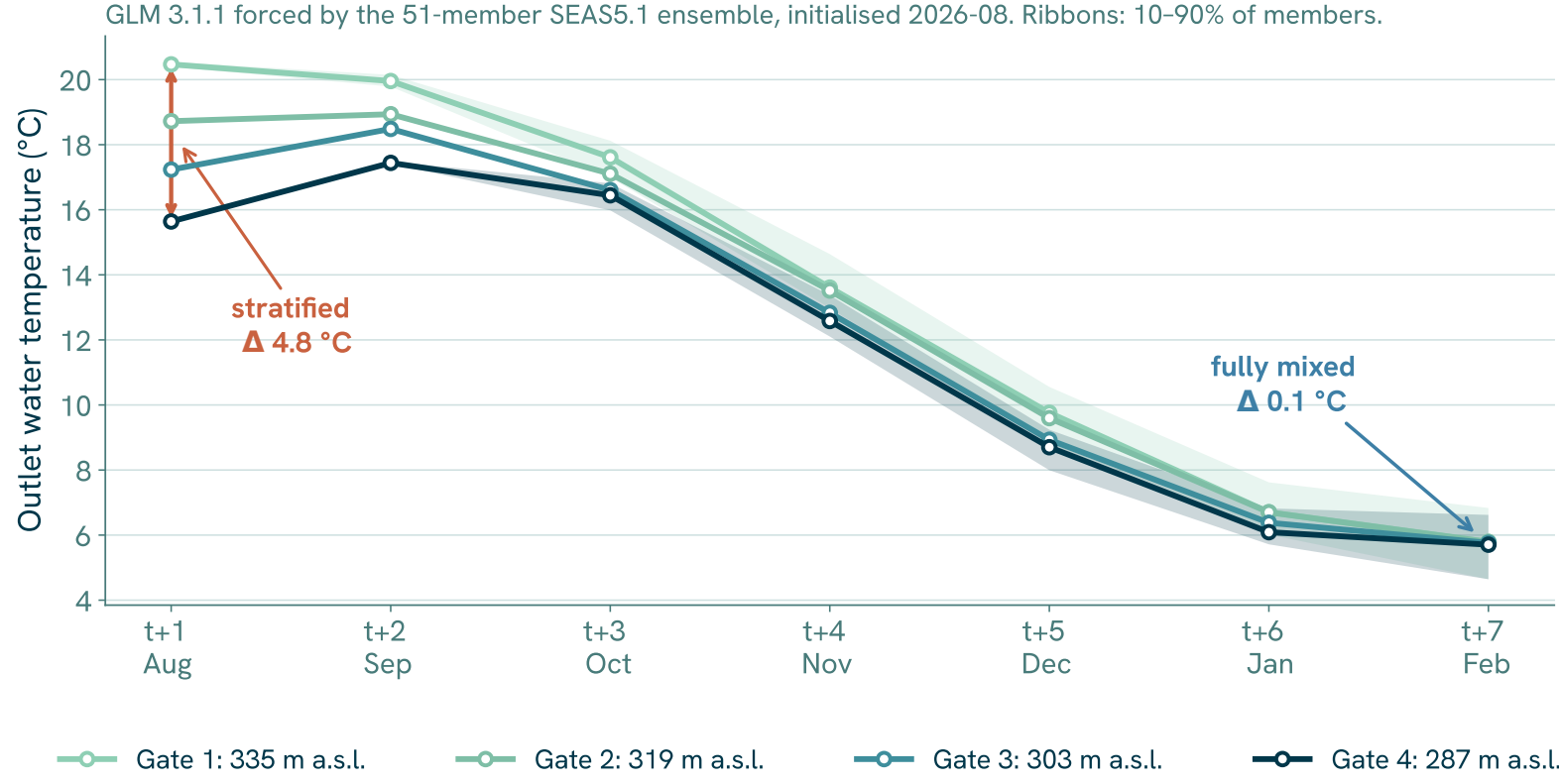


Fig. 3 The model carries the whole stratification cycle seven months out: **4.8 °C** between the shallowest and deepest gate in August, closing to **0.1 °C** at winter turnover.

Water quantity: Sau + Susqueda system storage

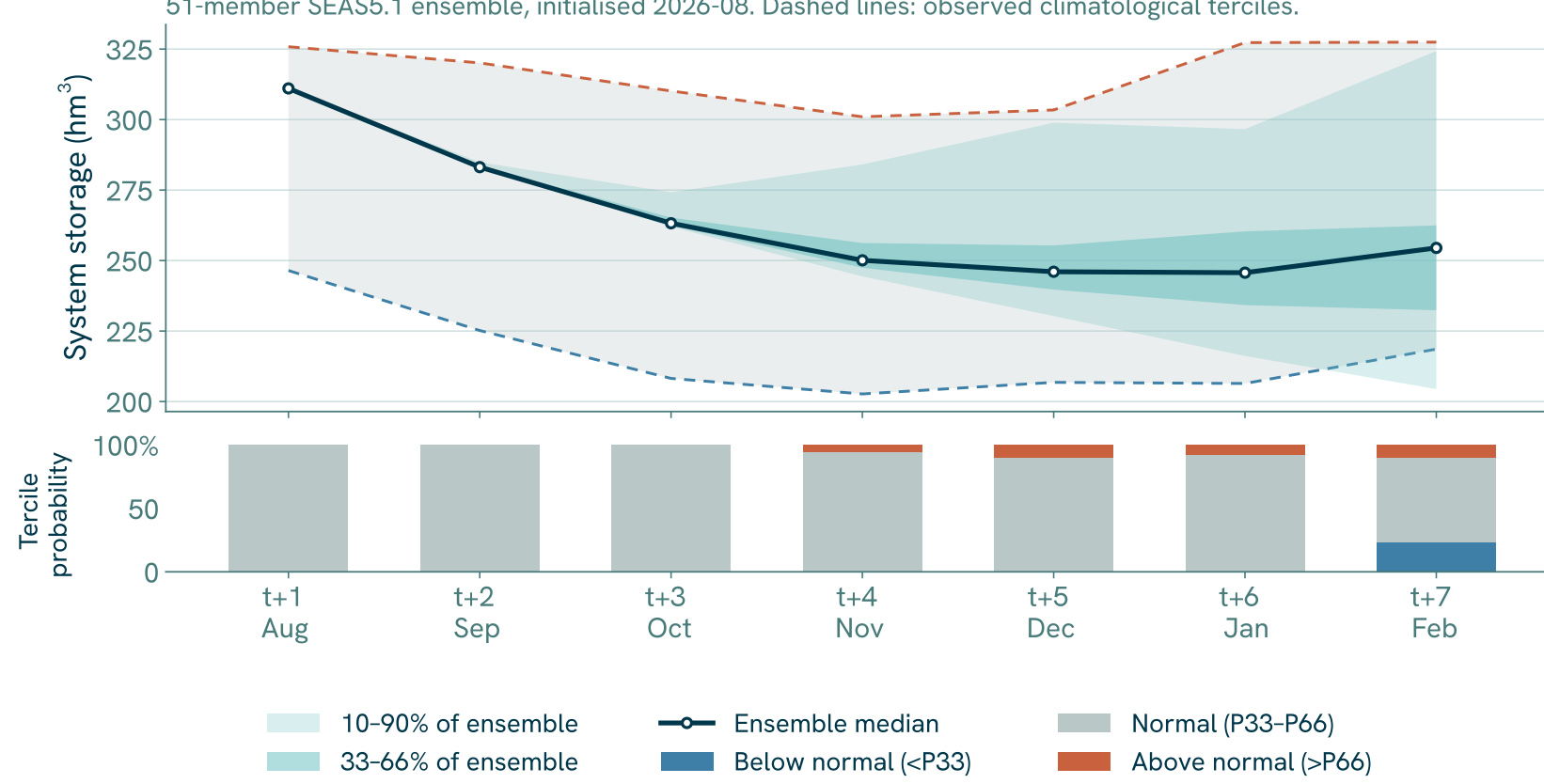


Fig. 2 Storage stays within the normal range through autumn, then picks up a **24% chance of falling below normal by February**, the earliest signal an operator gets.

Where the forecast actually has skill



Fig. 4 Verified against a 31-year hindcast, averaged over all twelve initialisation months. Only storage beats climatology across the full horizon.

• What we can and cannot predict

Storage is genuinely predictable

CRPSS 0.86 at one month and still 0.14 at seven: a reservoir integrates its own history, so the signal survives.

Inflow is not

Skill is positive only at t+1. Beyond that, climatology is as good as the forecast.

Gate temperature: right shape, no skill yet

The seasonal cycle is reproduced, but CRPSS stays below zero, it does not yet beat climatology.

PUBLISHED SKILL CRITERION

NSE > 0 & CRPSS ≥ 0.1

Sites that fail it are published as low- or no-skill rather than hidden.

Everything here is live: scan and use it

Open-source browser tools and dashboards. No installation, no account. All code at github.com/danielmerbet.



Seasonal forecast viewer

Interactive map of inflow and reservoir forecasts with ensemble spread and per-site skill.

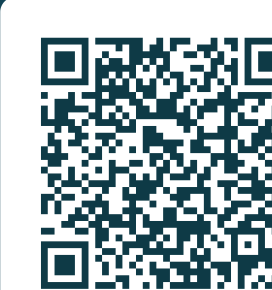
nereidaca.github.io/hydro_tool



Saline intrusion viewer

Forecast maps of salt-water intrusion in the coastal and estuarine reach.

nereidaca.github.io/salinintrusion_tool



fDOM forecast tool

Weather-driven forecasts of dissolved organic matter for drinking-water sources.

danielmerbet.github.io/fdom_weather_app



Catalunya groundwater map

Interactive map of groundwater status across Catalan aquifers.

danielmerbet.github.io/gw_catalunya

FUNDING & INSTITUTIONS

This work is part of the **NEREIDA** project, grant **RDI001/24/000044**, funded by the **Agència Catalana de l'Aigua (ACA)** under the R+D+I call **ACC/1362/2024**.



SELECTED BACKGROUND

- Mercado-Bettín, D. et al. (2021) Forecasting water temperature in lakes and reservoirs using seasonal climate prediction. *Water Research* 201, 117286.
- Jackson-Blake, L. A. et al. (2022) Opportunities for seasonal forecasting to support water management outside the tropics. *HESS* 26, 1389–1406.
- Clayer, F. et al. (2023) Sources of skill in lake temperature, discharge and ice-off seasonal forecasting tools. *HESS* 27, 1361–1381.
- Mercado-Bettín, D. et al. (2026) A machine learning approach to driver attribution of dissolved organic matter dynamics in two contrasting freshwater systems. *Biogeosciences* 23, 2661–2685.



All code, tools & publications
danielmerbet.github.io



Workshop · 31 Aug
Hybrid lake modelling
& forecasting, Lake Banyoles